

DBMS: ASSIGNMENT 4

Procedure and Function

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Case Study: Staff Payroll & Bonus Management System

Enter password: ****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 9

Server version: 8.0.45 MySQL Community Server - GPL

```
mysql> CREATE DATABASE StaffPayrollSystem;
```

Query OK, 1 row affected (0.04 sec)

```
mysql> USE StaffPayrollSystem;
```

Database changed

```
mysql> CREATE TABLE Staff (
```

```
-> Id INT PRIMARY KEY,
```

```
-> Name VARCHAR(50),
```

```
-> Salary INT,
```

```
-> Score INT
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO Staff VALUES
```

```
-> (1,'Adi Maitre',25000,5),
```

```
-> (2,'Kritika Chaudhari',22000,4);
```

Query OK, 2 rows affected (0.02 sec)

```
mysql> DELIMITER $$
```

```
mysql> CREATE PROCEDURE AddStaff(
```

```
-> IN p_id INT,
```

```
-> IN p_name VARCHAR(50),
```

```
-> IN p_salary INT,
```

```
-> IN p_score INT)
```

```
-> BEGIN
```

```
-> INSERT INTO Staff VALUES(p_id,p_name,p_salary,p_score);
```

```
-> END $$
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
```

```
mysql> CALL AddStaff(3,'Ritobrata Baksi',28000,5);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE UpdateSalary(
-> IN p_id INT,
-> IN new_salary INT)
-> BEGIN
-> UPDATE Staff SET Salary = new_salary WHERE Id = p_id;
-> END $$
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
```

```
mysql> CALL UpdateSalary(1,30000);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE FUNCTION CalculateBonus(p_id INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
-> DECLARE sal INT;
-> DECLARE score INT;
-> DECLARE bonus INT;
-> SELECT Salary, Score INTO sal, score FROM Staff WHERE Id = p_id;
-> IF score = 5 THEN
-> SET bonus = sal * 0.20;
-> ELSE
-> SET bonus = sal * 0.10;
-> END IF;
-> RETURN bonus;
-> END $$
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
```

```
mysql> SELECT CalculateBonus(1);
+-----+
| CalculateBonus(1) |
+-----+
| 6000             |
+-----+
```

```
mysql> SELECT * FROM Staff;
```

Id	Name	Salary	Score
1	Adi Maitre	30000	5
2	Kritika Chaudhari	22000	4
3	Ritobrata Baksi	28000	5

Project: Smart Digital Tax & Return Suite

```
mysql> CREATE DATABASE SmartTaxReturnSuite;
```

Query OK, 1 row affected (0.12 sec)

```
mysql> USE SmartTaxReturnSuite;
```

Database changed

```
mysql> CREATE TABLE user_login (
-> user_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
-> first_name VARCHAR(100) NOT NULL,
-> last_name VARCHAR(100) NOT NULL,
-> username VARCHAR(100) UNIQUE NOT NULL,
-> password_hash VARCHAR(255) NOT NULL,
-> role ENUM('admin','taxpayer') NOT NULL,
-> created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
```

Query OK, 0 rows affected (0.16 sec)

```
mysql> CREATE TABLE taxpayer_profile (
-> taxpayer_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
-> user_id BIGINT UNSIGNED UNIQUE,
-> pan_number VARCHAR(10) UNIQUE NOT NULL,
-> aadhaar CHAR(12),
-> city VARCHAR(100),
-> state VARCHAR(100),
-> FOREIGN KEY (user_id) REFERENCES user_login(user_id)
-> ON DELETE CASCADE ON UPDATE CASCADE
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> CREATE TABLE income_sources (
-> income_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
-> taxpayer_id BIGINT UNSIGNED,
-> source_type VARCHAR(100),
-> amount DECIMAL(15,2) DEFAULT 0.00,
-> financial_year VARCHAR(10),
-> FOREIGN KEY (taxpayer_id) REFERENCES taxpayer_profile(taxpayer_id)
```

-> ON DELETE CASCADE ON UPDATE CASCADE

->);

Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE deductions (

-> deduction_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

-> taxpayer_id BIGINT UNSIGNED,

-> deduction_type VARCHAR(100),

-> amount DECIMAL(15,2) DEFAULT 0.00,

-> financial_year VARCHAR(10),

-> FOREIGN KEY (taxpayer_id) REFERENCES taxpayer_profile(taxpayer_id)

-> ON DELETE CASCADE ON UPDATE CASCADE

->);

Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER \$\$

mysql> CREATE FUNCTION CalculateTaxableIncome(t_id BIGINT)

-> RETURNS DECIMAL(15,2)

-> DETERMINISTIC

-> BEGIN

-> DECLARE income DECIMAL(15,2);

-> DECLARE deduction DECIMAL(15,2);

-> SELECT SUM(amount) INTO income FROM income_sources WHERE taxpayer_id = t_id;

-> SELECT SUM(amount) INTO deduction FROM deductions WHERE taxpayer_id = t_id;

-> RETURN IFNULL(income,0) - IFNULL(deduction,0);

-> END \$\$

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;